

# Point

*From Number Theory to Algebraic Geometry and Logic*

## Session P0 — Set Theory, Order, and the Axiom of Choice

*Complete lecture notes with proofs, examples, and exercises*

### Abstract

This is the true session zero of the course — the one the first class actually needed. It builds the set-theoretic scaffolding everything else stands on: equivalence relations and their exact correspondence with partitions (the logic of every quotient to come:  $\mathbb{Z}/n$ ,  $G/N$ ,  $M \otimes N$ , sheafification); order relations (preorder, partial order, strict order) and the anatomy of extrema (minimum vs minimal, bounds, well-orders); and the axiomatic frame ZF with the **Axiom of Choice**, its equivalents (Well-Ordering, Zorn’s lemma — with the equivalence cycle proved or sketched), and its famous consequences and equivalents: bases of vector spaces, Tychonoff, Tarski’s theorem, spanning trees, and the precise tier of the Completeness Theorem. Zorn’s lemma is the engine of three existence theorems later in the course: Krull (S1), algebraic closures (S3), and Deligne (S58).

### Session plan (150 minutes)

| Time      | Block                                                                         |
|-----------|-------------------------------------------------------------------------------|
| 0:00–0:10 | Orientation: why set theory first; where P0 feeds (S1, S3, S2, S27, S58)      |
| 0:10–0:35 | §1 Equivalence relations and partitions (the logic of quotients)              |
| 0:35–1:00 | §2–§3 Orders: preorder/partial/strict; minima vs minimal; bounds; well-orders |
| 1:00–1:10 | <i>Break</i>                                                                  |
| 1:10–1:35 | §4 From Russell’s paradox to ZF                                               |
| 1:35–2:10 | §5 Axiom of Choice; the equivalence cycle; the list of equivalents            |
| 2:10–2:30 | Exercises / wrap-up: Zorn as the course’s existence engine                    |

### Conventions

A *relation* on  $X$  is a subset of  $X \times X$ ; we write  $x \sim y$  for  $(x, y) \in R$ . Orders are written  $\preceq$  (reflexive form) and  $\prec$  (strict form). “Poset” means partially ordered set. ZF denotes Zermelo–Fraenkel set theory without Choice; ZFC = ZF + AC.

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# 1 Equivalence Relations and Partitions

**Definition 1.1** (Equivalence relation; class; quotient). A relation  $\sim$  on  $X$  is an **equivalence relation** if it is reflexive ( $x \sim x$ ), symmetric ( $x \sim y \Rightarrow y \sim x$ ), and transitive. The **class** of  $x$  is  $[x] = \{y \in X : y \sim x\}$ ; the **quotient set** is  $X/\sim = \{[x] : x \in X\}$ .

**Definition 1.2** (Partition). A **partition** of  $X$  is a family  $\mathcal{P} \subseteq \mathcal{P}(X)$  of nonempty, pairwise disjoint sets whose union is  $X$ .

**Theorem 1.3** (Equivalence  $\leftrightarrow$  partition). (a) *If  $\sim$  is an equivalence relation on  $X$ , then  $X/\sim$  is a partition of  $X$ .*

(b) *If  $\mathcal{P}$  is a partition of  $X$ , then  $x \sim_{\mathcal{P}} y \iff \exists P \in \mathcal{P}, x, y \in P$  is an equivalence relation, and its classes are exactly the members of  $\mathcal{P}$ .*

*Hence equivalence relations on  $X$  and partitions of  $X$  are in canonical bijection.*

*Proof.* (a) Each  $x \in [x]$  by reflexivity, so the union is  $X$  and classes are nonempty. If  $z \in [x] \cap [y]$  then  $x \sim z$  and  $z \sim y$ , so  $x \sim y$ ; then for any  $u \in [x]$ :  $u \sim x \sim y$  gives  $u \in [y]$ , hence  $[x] \subseteq [y]$ ; symmetrically  $[y] \subseteq [x]$ . So distinct classes are disjoint. (b) Reflexive since  $\mathcal{P}$  covers  $X$ ; symmetric by definition; transitive because if  $x, y \in P$  and  $y, z \in Q$  then  $y \in P \cap Q$ , forcing  $P = Q$  by disjointness, so  $x, z \in P$ . The class of  $x$  is the unique  $P \ni x$ . The two passages invert each other.  $\square$

**Example 1.4** (Quotients everywhere). • Congruence mod  $n$  on  $\mathbb{Z}$ : classes  $[0], \dots, [n-1]$ ; quotient  $\mathbb{Z}/n\mathbb{Z}$  (S1).

- Construction of  $\mathbb{Q}$ : on  $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ ,  $(a, b) \sim (c, d) \iff ad = bc$ ; checking this is an equivalence relation *is* the construction of the rationals.
- Cosets  $G/H$  (P1), orbits of a group action (P1), stalks and sheafification (S26–S27), tensor products as quotients of free modules (S2): every quotient in the course is Theorem 1.3 at work — *identifying what should be the same point*.

**Remark 1.5** (Choosing representatives). A quotient map  $X \rightarrow X/\sim$  generally has no canonical section: picking one representative per class for infinitely many classes is exactly where the Axiom of Choice (§5) enters. Keep this in mind for S2 and S27.

## 2 Order Relations and Their Kin

**Definition 2.1** (Preorder, partial order, strict order, total order). A relation  $\preceq$  on  $P$  is a **preorder** if reflexive and transitive; a **partial order** if also antisymmetric ( $x \preceq y \preceq x \Rightarrow x = y$ ); **total/linear** if any two elements are comparable. A relation  $\prec$  is a **strict partial order** if irreflexive and transitive; a **strict total order** if additionally  $x \prec y$  or  $x = y$  or  $y \prec x$  for all  $x, y$ .

**Proposition 2.2** (The two forms of order correspond). *If  $\preceq$  is a partial order then  $x \prec y \iff (x \preceq y \wedge x \neq y)$  is a strict partial order; if  $\prec$  is a strict partial order then  $x \preceq y \iff (x \prec y \vee x = y)$  is a partial order. These passages are mutually inverse.*

*Proof.* Irreflexivity of  $\prec$  is immediate; transitivity: if  $x \prec y \prec z$  then  $x \preceq z$ , and  $x = z$  would give  $x \preceq y \preceq x$ , hence  $x = y$  by antisymmetry, contradicting  $x \prec y$ ; so  $x \neq z$ , i.e.  $x \prec z$ . The converse direction and the round-trips are similar case analyses.  $\square$

**Proposition 2.3** (A preorder is a partial order on its quotient). *If  $\preceq$  is a preorder on  $P$ , then  $x \sim y \iff (x \preceq y \wedge y \preceq x)$  is an equivalence relation, and  $[x] \preceq [y] \iff x \preceq y$  is a well-defined partial order on  $P/\sim$ . Thus §1 and §2 meet: every preorder becomes a poset after quotienting.*

*Proof.* Reflexivity/symmetry of  $\sim$  are built in; transitivity uses transitivity of  $\preceq$  twice. Well-definedness: if  $x \sim x'$  and  $x \preceq y$  then  $x' \preceq x \preceq y \preceq y'$ , so  $x' \preceq y'$ . Antisymmetry on classes is exactly the definition of  $\sim$ .  $\square$

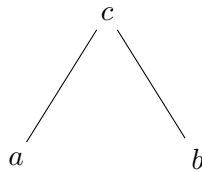
**Example 2.4.** •  $(\mathbb{N}, |)$  divisibility: partial, not total ( $2 \nmid 3$ ,  $3 \nmid 2$ ).

- $(\mathcal{P}(X), \subseteq)$ : partial; chains = nested families, antichains = pairwise disjoint (or incomparable) families.
- Preorder of functions  $f \preceq g \iff f = O(g)$  (analysis): not antisymmetric — quotienting gives the poset of growth rates (Proposition 2.3).
- Lexicographic order on  $\mathbb{R}^2$ : total.

### 3 Extrema, Bounds, and Well-Orders

**Definition 3.1** (Bounds and extrema). In a poset  $(P, \preceq)$  with  $S \subseteq P$ :  $u$  is an **upper bound** of  $S$  if  $s \preceq u$  for all  $s \in S$  (lower bound dually); the **supremum** is the least upper bound.  $m \in S$  is the **minimum** of  $S$  if  $m \preceq s$  for all  $s \in S$ ;  $m$  is **minimal** if no  $s \in S$  satisfies  $s \prec m$ . Maximum/maximal are dual.

**Example 3.2** (Minimum  $\neq$  minimal). In  $P = \{a, b, c\}$  with  $a \prec c$ ,  $b \prec c$  and  $a, b$  incomparable (Figure 1): the minimal elements are  $a$  and  $b$  (two of them!), there is *no* minimum;  $c$  is both maximum and maximal; the set  $\{a, b\}$  has upper bound  $c$  (its supremum) but no maximum. In  $(\mathbb{N}_{\geq 2}, |)$ : minimal elements are the primes; the minimum is 2? no — 2 does not divide 3: there is no minimum, only minima.



two minimal elements, no minimum;  $c = \text{maximum} = \text{supremum of } \{a, b\}$

Figure 1: Hasse diagram: minimal vs minimum, maximal vs maximum, bounds vs extrema.

**Definition 3.3** (Well-order). A total order  $\preceq$  on  $W$  is a **well-order** if every nonempty  $S \subseteq W$  has a minimum. Examples:  $(\mathbb{N}, \leq)$ ;  $(\mathbb{Z}, \leq)$  and  $(\mathbb{R}, \leq)$  are *not*; every ordinal is.

**Proposition 3.4.** A well-order has no infinite strictly decreasing chain  $x_1 \succ x_2 \succ \dots$ ; conversely, a total order with no infinite decreasing chain is a well-order provided countable choice is available (over ZF the forward direction is unconditional).

*Proof.* If  $x_1 \succ x_2 \succ \dots$  exists then  $\{x_i : i \in \mathbb{N}\}$  is nonempty with no minimum. Conversely, if  $S \neq \emptyset$  has no minimum, pick  $x_1 \in S$ ; having  $x_n$ , the set  $\{s \in S : s \prec x_n\}$  is nonempty (else  $x_n$  were minimal=minimum), so choose  $x_{n+1}$  in it — producing a decreasing chain. (The recursive choices use dependent choice; note where.)  $\square$

### 4 From Russell's Paradox to ZF

**Proposition 4.1** (Russell). There is no set  $R$  with  $R = \{x : x \notin x\}$ .

*Proof.* If  $R$  were a set, then  $R \in R \iff R \notin R$  by definition — contradiction. Hence “the set of all sets” cannot exist: comprehension must be restricted.  $\square$

**Definition 4.2** (The ZF axioms, informally). Language: only  $\in$  and  $=$ . Axioms: **Extensionality** (same elements  $\Rightarrow$  same set); **Pairing**  $\{x, y\}$ ; **Union**  $\bigcup \mathcal{F}$ ; **Power set**  $\mathcal{P}(X)$ ; **Infinity** (an inductive set exists); **Separation** (from a set, carve out by any formula — the safe replacement of naive comprehension); **Replacement** (images of sets under definable functions are sets); **Foundation** (no infinite  $\in$ -chains; every nonempty set has an  $\in$ -minimal element). **Choice** is *not* among them — it is the guest of §5.

**Remark 4.3** (What ZF cannot do alone). ZF proves each  $X_i \neq \emptyset$  has elements, but not that an *infinite family* admits a simultaneous choice: Russell’s socks (no distinguishing property) vs shoes (left/right gives a rule). Everything in §5 measures exactly this gap.

## 5 The Axiom of Choice and Its Equivalents

**Definition 5.1** (AC, WO, Zorn). **AC**: every family  $\{X_i\}_{i \in I}$  of nonempty sets has a choice function  $c : I \rightarrow \bigcup_i X_i$  with  $c(i) \in X_i$ ; equivalently  $\prod_i X_i \neq \emptyset$ . **WO (Well-Ordering Principle)**: every set admits a well-order. **Zorn (ZL)**: if  $(P, \preceq) \neq \emptyset$  and every chain has an upper bound in  $P$ , then  $P$  has a maximal element.

**Proposition 5.2** ( $\text{WO} \Rightarrow \text{AC}$ , in ZF).

*Proof.* Well-order  $W = \bigcup_i X_i$ . For each  $i$ ,  $X_i$  is a nonempty subset of  $W$ , hence has a  $W$ -least element; set  $c(i) = \min_W X_i$ . This is a choice function — no choice used in the proof.  $\square$

**Proposition 5.3** ( $\text{Zorn} \Rightarrow \text{AC}$ , in ZF).

*Proof.* Let  $\mathcal{F}$  be the set of partial choice functions:  $f \in \mathcal{F}$  iff  $f$  is a function with  $\text{dom}(f) \subseteq I$  and  $f(i) \in X_i$ . Order  $\mathcal{F}$  by extension;  $\mathcal{F} \neq \emptyset$  (empty function). A chain  $\mathcal{C}$  has upper bound  $\bigcup \mathcal{C}$  (union of compatible functions is a partial choice function). Zorn gives maximal  $f$ . If  $\text{dom}(f) \neq I$ , pick  $i \notin \text{dom}(f)$  and  $x \in X_i$  (a *single* choice, legal in ZF) and extend  $f$  by  $i \mapsto x$  — contradicting maximality. Hence  $\text{dom}(f) = I$ .  $\square$

**Proposition 5.4** ( $\text{AC} \Rightarrow \text{WO}$  and  $\text{AC} \Rightarrow \text{Zorn}$ : sketches).  $\text{AC} \Rightarrow \text{WO}$ : fix a choice function  $c$  on  $\mathcal{P}(X) \setminus \{\emptyset\}$ ; by transfinite recursion set  $x_\alpha = c(X \setminus \{x_\beta : \beta < \alpha\})$ ; the recursion must stop (no injection from the class of ordinals into a set — Hartogs), and at the stopping stage  $X$  is exhausted, so  $\alpha \mapsto x_\alpha$  well-orders  $X$ .  $\text{AC} \Rightarrow \text{Zorn}$ : assuming no maximal element, recursively build a strictly increasing chain  $x_\alpha$  by choosing at each stage a strict upper bound of  $\{x_\beta : \beta < \alpha\}$  (which exists since chains have upper bounds and no upper bound is maximal); again Hartogs bounds the length — contradiction.

**Corollary 5.5.** Over ZF, the three statements AC, WO, ZL are equivalent (Propositions 5.2, 5.3, 5.4); so are Hausdorff’s Maximality Principle and Tukey’s lemma.

**Theorem 5.6** (The list of equivalents and near-equivalents). Over ZF, the following are *equivalent* to AC:

- (1) Every vector space has a basis (Blass, 1984).
- (2) Tychonoff’s theorem: arbitrary products of compact spaces are compact.
- (3) Tarski’s theorem: for every infinite  $A$ ,  $|A \times A| = |A|$ .
- (4) Every connected graph has a spanning tree (forward direction: Zorn on forests ordered by inclusion).

(5) Every surjection has a section (a reformulation of AC itself).

And the following live in strictly weaker tiers (important to know, easy to overstate): the **Boolean Prime Ideal / Ultrafilter Lemma** and the general **Gödel Completeness Theorem** (equivalent to BPI; the countable-language case is provable in ZF); **Hahn–Banach**; and mere consequences of AC such as the existence of non-measurable sets.

**Remark 5.7** (Where the course spends Zorn). Krull’s theorem (S1: maximal ideals exist)  $\rightarrow$  algebraic closures (S3)  $\rightarrow$  Deligne’s theorem (S58: coherent topoi have enough points). Three existence theorems, one engine — and each is a “points exist” statement, fitting the course’s theme perfectly.

## 6 Exercises

**Exercise 1.** Prove both directions of Theorem 1.3 for: congruence mod  $n$  on  $\mathbb{Z}$ ; and  $x \sim y \iff |x| = |y|$  on  $\mathbb{R}$  (describe the partition geometrically).

**Exercise 2.** Carry out the construction of  $\mathbb{Q}$  from  $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ : verify  $(a, b) \sim (c, d) \iff ad = bc$  is an equivalence relation, and that  $[(a, b)] + [(c, d)] = [(ad + bc, bd)]$  is well defined.

**Exercise 3.** Show  $f \preceq g \iff f = O(g)$  is a preorder on positive functions  $\mathbb{N} \rightarrow \mathbb{R}_{>0}$  but not a partial order; describe the quotient poset (growth rates) and identify its minimum.

**Exercise 4.** In  $(\mathcal{P}(X) \setminus \{\emptyset\}, \subseteq)$  with  $|X| \geq 2$ : find minimal and maximal elements; does a minimum exist? Repeat for  $(\mathbb{N}_{\geq 2}, |)$ .

**Exercise 5.** Prove Proposition 2.2 in full (all case analyses), and show lexicographic order on  $\mathbb{R}^2$  is a strict total order.

**Exercise 6.** Prove the forward direction of Proposition 3.4 in ZF, and pinpoint exactly where the converse uses dependent choice; give a poset with no decreasing chain that is not well-ordered.

**Exercise 7.** Prove Proposition 5.3 in full detail (verify  $\bigcup \mathcal{C}$  is a function and a partial choice function; verify maximality forces full domain).

**Exercise 8.** Use Zorn to prove Krull’s theorem: every proper ideal of a commutative ring with  $1 \neq 0$  lies in a maximal ideal (this is S1’s engine — prove it here first).

**Exercise 9.** Use Zorn to prove every vector space has a basis (order the linearly independent subsets by inclusion); identify where the chain upper bound uses unions.

**Exercise 10.** Prove Tychonoff’s theorem for *finite* products in ZF by induction, and locate the exact step where the general proof must invoke choice (ultrafilters or subbases).

**Exercise 11.** Show  $AC \iff$  every surjection  $p : E \rightarrow B$  has a section  $s : B \rightarrow E$  ( $p \circ s = \text{id}_B$ ); interpret sections as “continuous choices of points of fibers” and compare with Remark 1.4.

**Exercise 12.** Explain why Russell’s paradox does not contradict Separation: for a set  $A$ , show  $\{x \in A : x \notin x\} \notin A$ ; conclude no universal set exists in ZF.

## Where This Session Feeds the Course

- **Depends on:** nothing — this is session zero.
- **Feeds:** S1 (Krull via Zorn; quotients  $R/I$ ); S2 (tensor as quotient; choosing representatives); S3 (algebraic closure via Zorn); S27 (sheafification classes); S34/S16 (orderings of finite subextensions for inverse limits); S58 (Deligne via Zorn); P1 (cosets as partitions, Exercise 12 preview).

- **Unifying thread:** quotient by equivalence = identifying points that must agree; order = the scaffolding on which Zorn climbs; Zorn = the engine that guarantees the *existence* of the points (maximal ideals, closures, models) the course studies.

## References for this session

- [1] Halmos, P.R. *Naive Set Theory* (the classic first read).
- [2] Enderton, H.B. *Elements of Set Theory* (proofs of the equivalence cycle).
- [3] Jech, T. *The Axiom of Choice / Set Theory* (the definitive treatment).
- [4] Howard, P. & Rubin, J. *Consequences of the Axiom of Choice* (the list, with tiers).
- [5] Davey, B. & Priestley, H. *Introduction to Lattices and Order* (order theory).

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*End of Session P0 — next: Session P1, Group Theory: Part I*